

IELTS READING TEST

PASSAGE 1

YAWNING

How and why we yawn still presents problems for researchers in an area which has only recently been opened up to study

When Robert R. Provine began studying yawning in the 1960s, it was difficult for him to convince research students of the merits of 'yawning science'. Although it may appear quirky to some, Provine's decision to study yawning was a logical extension of his research in developmental neuroscience.

The verb 'to yawn' is derived from the Old English *ganien* or *ginian*, meaning to gape or open wide. But in addition to gaping jaws, yawning has significant features that are easy to observe and analyse. Provine 'collected' yawns to study by using a variation of the contagion response*. He asked people to 'think about yawning' and, once they began to yawn to depress a button and that would record from the start of the yawn to the exhalation at its end.

Provine's early discoveries can be summarized as follows: the yawn is highly stereotyped but not invariant in its duration and form. It is an excellent example of the instinctive 'fixed action pattern' of classical animal-behavior study, or ethology. It is not a reflex (short-duration, rapid, proportional response to a simple stimulus), but, once started, a yawn progresses with the inevitability of a sneeze. The standard yawn runs its course over about six seconds on average, but its duration can range from about three seconds to much longer than the average. There are no half-yawns: this is an example of the typical intensity of fixed action patterns and a reason why you cannot stifle yawns. Just like a cough, yawns can come in bouts with a highly variable inter-yawn interval, which is generally about 68 seconds but rarely more than 70. There is no relation between yawn frequency and duration: producers of short or long yawns do not compensate by yawning more or less often. Furthermore, Provine's hypotheses about the form and function of yawning can be tested by three informative yawn variants which can be used to look at the roles of the nose, the mouth and the jaws.

i) The closed nose yawn

Subjects are asked to pinch their nose closed when they feel themselves start to yawn. Most subjects report being able to perform perfectly normal closed nose yawns. This

indicates that the inhalation at the onset of a yawn, and the exhalation at its end, need not involve the nostrils – the mouth provides a sufficient airway.

ii) The clenched teeth yawn

Subjects are asked to clench their teeth when they feel themselves start to yawn but allow themselves to inhale normally through their open lips and clenched teeth. This variant gives one the sensation of being stuck mid-yawn. This shows that gaping of the jaws is an essential component of the fixed action pattern of the yawn, and unless it is accomplished, the program (or pattern) will not run to completion. The yawn is also shown to be more than a deep breath, because, unlike normal breathing, inhalation and exhalation cannot be performed so well through the clenched teeth as through the nose.

iii) The nose yawn

This variant tests the adequacy of the nasal airway to sustain a yawn. Unlike normal breathing, which can be performed equally well through mouth or nose, yawning is impossible via nasal inhalation alone. As with the clenched teeth yawn, the nose yawn provides the unfulfilling sensation of being stuck in mid-yawn. Exhalation, on the other hand, can be accomplished equally well through nose or mouth. Through thin methodology Provine demonstrated that inhalation through the oral airway and the gaping of jaws are necessary for normal yawns. The motor program for yawning will not run to completion without feedback that these parts of the program have been accomplished.

But yawning is a powerful, generalized movement that involves much more than airway manoeuvres and jaw-gaping. When yawning you also stretch your facial muscles, tilt your head back, narrow or close your eyes, produce tears, salivate, open the Eustachian tubes of your middle ear and perform many other, yet unspecified, cardiovascular and respiratory acts. Perhaps the yawn shares components with other behaviour. For example, in the yawn a kind of 'slow sneeze' or is the sneeze a 'fast yawn'? Both share common respiratory and other features including jaw gaping, eye closing and head tilting.

Yawning and stretching share properties and may be performed together as parts of a global motor complex. Studies by J I p deVries et al. in the early 1980s, charting movement in the developing foet US using ultrasound, observed a link between yawning and stretching. The most extraordinary demonstration of the yawn-stretch

linkage occurs in many people paralyzed on one side of their body because of brain damage caused by a stroke, the prominent British neurologist Sir Francis Walshe noted in 1923 that when these people yawn, they are startled and mystified to observe that their otherwise paralyzed arm rises and flexes automatically in what neurologists term an 'associated response'. Yawning apparently activates undamaged, unconsciously controlled connections between the brain and the motor system, causing the paralyzed limb to move. It is not known whether the associated response is a positive prognosis for recovery, nor whether yawning is therapeutic for prevention of muscular deterioration.

Provine speculated that, in general, yawning may have many functions, and selecting a single function from the available options may be an unrealistic goal. Yawning appears to be associated with a change of behavioral state, switching from one activity to another. Yawning is also a reminder that ancient and unconscious behavior linking US to the animal world lurks beneath the veneer of culture, rationality and language.

Questions 1-6

Complete the summary below using the list of words, A-K, below

Write the correct letter, A-K, in boxes 1-6 on your answer sheet.

Provine's early findings on yawns

Through his observation of yawns, Province was able to confirm that **1** _____ do not exist.

Just like a **2** _____, yawns cannot be interrupted after they have begun. This is because yawns occur as a **3** _____ rather than a stimulus response as was previously thought.

In measuring the time taken to yawn, Province found that a typical yawn lasts about **4** _____. He also found that it is a common for people to yawn a number of times in quick succession with the yawns usually being around **5** _____ apart.

When studying whether length and rate were connected, Province concluded that people who yawn less do not necessarily produce **6** _____ to make up for this.

- | | | |
|-------------------------------|----------------------|-------------------------|
| A form and function | E 68 seconds | I short duration |
| B long yawns | F short yawns | J 6 seconds |
| C 3 seconds | G reflex | K half-yawns |
| D fixed action pattern | H sneeze | |

Questions 7-11

Choose the correct letter, A, B, C or D.

Write the correct letter in boxes 7-11 on your answer sheet.

7. What did Provine conclude from his 'closed nose yawn' experiment?

- A. Ending a yawn requires use of the nostrils.
- B. You can yawn without breathing through your nose
- C. Breathing through the nose produces a silent yawn.
- D. The role of the nose in yawning needs further investigation.

8. Provine's clenched teeth yawn's experiment shows that

- A. yawning is unconnected with fatigue.
- B. a yawn is the equivalent of a deep intake of breath.
- C. you have to be able to open your mouth wide to yawn.
- D. breathing with the teeth together is as efficient as through the nose.

9. The nose yawn experiment was used to test whether yawning

- A. can be stopped after it has started
- B. is the result of motor programming
- C. involves both inhalation and exhalation.
- D. can be accomplished only through the nose.

10. In people paralyzed on one side because of brain damage

- A. yawning may involve only one side of the face.
- B. the yawning response indicates that recovery is likely
- C. movement in paralysed arm is stimulated by yawning
- D. yawning can be used as an example to prevent muscle wasting.

11. In the last paragraph, the writer concludes that

- A. yawning is a sign of boredom.
- B. we yawn in spite of the development of our species
- C. yawning is a more passive activity than we imagine
- D. we are stimulated to yawn when our brain activity is low

Questions 12-14

Do the following statements agree with the claims of the writer in Reading Passage?

In boxes 12-14 on your answer sheet, write

YES if the statement agrees with the views of the writer

NO if the statement contradicts the views of the writer

NOT GIVEN if it is impossible to say what the writer thinks about this

12 Research students were initially reluctant to appreciate the value of Provine's studies. ☐

13 When fetuses yawn and stretch they are learning how to control movement. ☐

14 According to Provine, referring to only one function is probably inadequate to explain why people yawn. ☐

PASSAGE 2

Water Filter

A. An ingenious invention is set to bring clean water to the third world, and while the science may be cutting edge, the materials are extremely down to earth. A handful of clay, yesterday's coffee grounds and some cow manure are the ingredients that could bring clean, safe drinking water to much of the third world.

B. The simple new technology, developed by ANU materials scientist Mr. Tony Flynn, allows water filters to be made from commonly available materials and fired on the ground using cow manure as the source of heat, without the need for a kiln. The filters have been tested and shown to remove common pathogens (disease-producing organisms) including E-coli. Unlike other water filtering devices, the filters are simple and inexpensive to make. "They are very simple to explain and demonstrate and can be made by anyone, anywhere," says Mr. Flynn. "They don't require any western technology. All you need is terracotta clay, a compliant cow and a match."

C. The production of the filters is extremely simple. Take a handful of dry, crushed clay, mix it with a handful of organic material, such as used tea leaves, coffee grounds or rice hulls, add enough water to make a stiff biscuit-like mixture and form a cylindrical pot that has one end closed, then dry it in the sun. According to Mr. Flynn, used coffee grounds have given the best results to date. Next, surround the pots with straw; put them in a mound of cow manure, light the straw and then top up the burning manure as required. In less than 60 minutes the filters are finished. The walls of the finished pot should be about as thick as an adult's index. The properties of cow manure are vital as the fuel can reach a temperature of 700 degrees in half an hour and will be up to 950 degrees after another 20 to 30 minutes. The manure makes a good fuel because it is very high in organic material that burns readily and quickly; the manure has to be dry and is best used exactly as found in the field, there is no need to break it up or process it any further.

D. "A potter's kiln is an expensive item and can take up to four or five hours to get up to 800 degrees. It needs expensive or scarce fuel, such as gas or wood to heat it and experience to run it. With no technology, no insulation and nothing other than a pile of cow manure and a match, none of these restrictions apply," Mr. Flynn says.

E. It is also helpful that, like terracotta clay and organic material, cow dung is freely available across the developing world. "A cow is a natural fuel factory. My understanding is that cow dung as a fuel would be pretty much the same wherever you would find it." Just as using manure as a fuel for domestic uses is not a new idea, the porosity of clay is something that potters have known about for years, and something that as a former ceramics lecturer in the ANU School of Art, Mr. Flynn is well aware of. The difference is that rather than viewing the porous nature of the material as a problem after all not many people want a pot that won't hold water -his filters capitalize on this property.

F. Other commercial ceramic filters do exist, but, even if available, with prices starting at US\$5 each, they are often outside the budgets of most people in the developing world. The filtration process is simple, but effective. The basic principle is that there are passages through the filter that are wide enough for water droplets to pass through, but too narrow for pathogens. Tests with the deadly E-coli bacterium have seen the filters remove 94.6 to 99.8 percent of the pathogen well within safe level. Using only one filter it takes two hours to filter a litre of water. The use of organic material, which burns away leaving cavities after firing, helps produce the structure in which pathogens will become trapped. It overcomes the potential problems of finer clays that may not let water through and also means that cracks are soon halted. And like clay and cow dung, it is universally available.

G. The invention was born out of a World Vision project involving the Manatuto community in East Timor. The charity wanted to help set up a small industry manufacturing water filters, but initial research found the local clay to be too fine a problem solved by the addition of organic material. While the problems of producing a working ceramic filter in East Timor were overcome, the solution was kiln-based and particular to that community's materials and couldn't be applied elsewhere. Manure firing, with no requirement for a kiln, has made this zero technology approach available anywhere it is needed. With all the components being widely available, Mr. Flynn says there is no reason the technology couldn't be applied throughout the developing world, and with no plans to patent his idea, there will be no legal obstacles to it being adopted in any community that needs it. "Everyone has a right to clean water, these filters have the potential to enable anyone in the world to drink water safely," says Mr. Flynn.

Questions 15-19

Complete the flow chart, using **NO MORE THAN TWO WORDS** from the Reading Passage for each answer.

Write your answers in boxes 14-19 on your answer sheet.

Guide to Making Water Filters

Step one: combination of clay and organic material, with sufficient **15** _____ to create a thick mixture



sun dried

Step two: pack **16** _____ around the cylinders

Place them in **17** _____



for firing (maximum temperature: **18**_____)

filter being baked in under **19**_____

Questions 20-23

Do the following statements agree with the information given in Reading Passage 2?

In boxes 20-23 on your answer sheet, write

TRUE if the statement is true

FALSE if the statement is false

NOT GIVEN if the information is not given in the passage

20 It takes half an hour for the manure to reach 950 degrees. ☐

21 Clay was initially found to be unsuitable for filter making. ☐

22 Coffee grounds are twice as effective as other materials. ☐

23 E-coli is the most difficult bacteria to combat. ☐

Questions 24-26

Choose the correct letter, A, B, C or D.

Write your answers in boxes 24-26 on your answer sheet.

24 When making the pot, the thickness of the wall _____

- A.** is large enough to let the pathogens to pass.
- B.** varied according to the temperature of the fuel.
- C.** should be the same as an adult's forefinger.
- D.** is not mentioned by Mr. Flynn.

25 What is true about the charity, it _____

- A.** failed in searching the appropriate materials.
- B.** thought a kiln is essential.
- C.** found that the local clay are good
- D.** intended to build a filter production factory.

26 Mr. Flynn's design is purposely not being patented_____

- A.** because he hopes it can be freely used around the world
- B.** because he doesn't think the technology is perfect enough
- C.** because there are some legal obstacles
- D.** because the design has already been applied thoroughly

PASSAGE 3

Memory Decoding

A. Try this memory test: Study each face and compose a vivid image for the person's first and last name. Rose Leo, for example, could be a rosebud and a lion. Fill in the blanks on the next page. The Examinations School at Oxford University is an austere building of oak-paneled rooms, large Gothic windows, and looming portraits of eminent dukes and earls. It is where generations of Oxford students have tested their memory on final exams, and it is where, last August, 34 contestants gathered at the World Memory Championships to be examined in an entirely different manner. In timed trials, contestants were challenged to look at and then recite a two-page poem, memorize rows of 40-digit numbers, recall the names of 110 people after looking at their photographs, and perform seven other feats of extraordinary retention. Some tests took just a few minutes; others lasted hours. In the 14 years since the World Memory Championships was founded, no one has memorized the order of a shuffled deck of playing cards in less than 30 seconds. That nice round number has become the four-minute mile of competitive memory; a benchmark that the world's best "mental athletes," as some of them like to be called, are closing in on. Most contestants claim to have just average memories, and scientific testing confirms that they're not just being modest. Their feats are based on tricks that capitalize on how the human brain encodes information. Anyone can learn them.

B. Psychologists Elizabeth Valentine and John Wilding, authors of the monograph *Superior Memory*, recently teamed up with Eleanor Maguire, a neuroscientist at University College London to study eight people, including Karsten, who had finished near the top of the World Memory Championships. They wondered if the contestants' brains were different in some way. The researchers put the competitors and a group of control subjects into an MRI machine and asked them to perform several different memory tests while their brains were being

scanned. When it came to memorizing sequences of three-digit numbers, the difference between the memory contestants and the control subjects was, as expected, immense. However, when they were shown photographs of magnified snowflakes, images that the competitors had never tried to memorize before, the champions did no better than the control group. When the researchers analyzed the brain scans, they found that the memory champs were activating some brain regions that were different from those the control subjects were using. These regions, which included the right posterior hippocampus, are known to be involved in visual memory and spatial navigation.

C. It might seem odd that the memory contestants would use to visual imagery and spatial navigation to remember numbers, but the activity makes sense when their techniques are revealed. Cooke, a 23-year-old cognitive-science graduate student with a shoulder-length mop of curly hair, is a grand master of brain storage. He can memorize the order of 10 decks of playing cards in less than an hour or one deck of cards in less than a minute. He is closing in on the 30-second deck. In the Lamb and Flag, Cooke pulled out a deck of cards and shuffled it. He held up three cards the 7 of spades, the queen of clubs, and the 10 of spades. He pointed at a fireplace and said. "Destiny's Child is whacking Franz Schubert with handbags." The next three cards were the king of hearts, the king of spades, and the jack of clubs. He ran over to the bar and announced, "Admiral Lord Nelson is holding a guitar upside down over there." By now, everyone in the pub had begun to gawk. Forty-six cards and a few minutes later, Cooke ended up outside the Lamb and Flag, where he proceeded to reel off the deck's order flawlessly.

D. How did he do it? Cooke has already memorized a specific person, verb, and object that he associates with each card in the deck. For example, for the 7 of spades, the person (or, in this case, persons) is always the singing group Destiny's Child, the action is surviving a storm, and the image is a dinghy. The queen of clubs is always his friend Henrietta, the action is thwacking with a handbag, and the image is of wardrobes filled with designer clothes. When Cooke commits a deck to memory, he does it three cards at a time. Every three-card group forms a single image of a person doing something to an object. The first card in the triplet becomes the person, the second the verb, the third the object. He then places those images along a specific familiar route, such as the one he took through the Lamb and Flag. In competitions, he uses an imaginary route that He has designed to be as smooth and downhill

as possible. When it comes time to recall, Cooke takes a mental walk along his route and translates the images into cards. That's why the MRIs of the memory contestants showed activation in the brain areas associated with visual imagery and spatial navigation.

E. The more resonant the images are, the more difficult they are to forget. But even meaningful information is hard to remember when there's a lot of it. That's why competitive memorizers place their images along an imaginary route. That technique, known as the loci method, reportedly originated in 477 B.C. with the Greek poet Simonides of Ceos. Simonides was the sole survivor of a roof collapse that killed all the other guests at a royal banquet. The bodies were mangled beyond recognition, but Simonides was able to reconstruct the guest list by closing his eyes and recalling each individual around the dinner table. What he had discovered was that our brains are exceptionally good at remembering images and spatial information. Evolutionary psychologists have offered an explanation: Presumably our ancestors found it important to recall where they found their last meal or the way back to the cave. After Simonides' discovery, the loci method popular across ancient Greece as a trick for memorizing speeches and texts. Aristotle wrote about it, and later a number of treatises on the art memory were published in Rome. Before printed books, the art of memory was considered a staple or classical education on a par with grammar, logic and rhetoric.

F. The most famous of the naturals was the Russian journalist S. V. Shereshevski, who could recall long lists of numbers memorized decades earlier, as well as poems, strings of nonsense syllables, and just about anything else he was asked to remember. "The capacity of his memory had no distinct limits." wrote Alexander Luria, the Russian psychologist who studied Shereshevski from the 1920s to the 1950s. Shereshevski also had synesthesia, a rare condition in which the senses become intertwined. For example, every number may be associated with a color or every word with a taste. Synesthetic reactions evoke a response in more areas of the brain, making memory easier. They also create problems. "If I read when I eat, I have a hard time understanding what I'm reading the taste of the food drowns out the sense." Shereshevski told Luria.

G. K. Anders Ericsson, a Swedish-born psychologist at Florida State University, thinks anyone can acquire Shereshevski's skills. He cites an experiment with S. F., an undergraduate who was paid to take a standard test of memory called the digit span for one hour a day, two or

three days a week. When he started, he could hold, like most people, only about seven digits in his head at any given time (conveniently, the length of a phone number). Over two years, S.F. completed 250 hours of testing. By then, he had stretched his digit span from 7 to more than 80. He had developed his own strategy for remembering based on his own experience as a competitive runner: He associated strings of random numbers with track times. For example 3,492 was remembered as "3 minutes and 49 point 2 seconds, near world-record mile time." The study of S. F. led Ericsson to believe that innately superior memory doesn't exist at all. When he reviewed original case studies of naturals, he found that exceptional memorizers were using techniques sometimes without realizing it and lots of practice. Often, exceptional memory was only for a single type of material, like digits. "If we look at some of these memory tasks, they're the kind of thing most people don't even waste one hour practicing, but if they wasted 50 hours, they'd be exceptional at it," Ericsson says. It would be remarkable, he adds, to find a person who is exceptional across a number of tasks. I don't think that there's any compelling evidence that there are such people."

Questions 27-30

The reading Passage has seven paragraphs A-G.

Which paragraph contains the following information?

Write the correct letter A-G in boxes 27-30 on your answer sheet.

NB You may use any letter more than once.

27 The reason why competence of super memory is significant in academic settings

28 An example of extraordinary recalling by applying orders. designed images

29 for good Scientific experiment into Biological explanation for good memory

30 A depiction of natural ability of memory

Questions 31-35

Complete the following summary of the paragraphs of Reading Passage, using **NO MORE THAN THREE WORDS** from the Reading Passage for each answer. Write your answers in boxes 31-35 on your answer sheet.

In World Memory Championships. **31**_____ was a limitation time for participants to remember a deck of cards. A man called Ed Cooke in a pub, spoke a string of odd words when he held 7 of the spades (one of the three cards) was remembered as he encoded it to a **32**_____. The superior memory skill can be traced back to Ancient Greece, the strategy was called **33**_____ journalist had a similar mental condition named **34**_____ Russian which combined number with other senses, but Ericsson suggested that fantastic memory could be achieved by numerous **35**_____.

Questions 36-40

Do the following statements agree with the information given in Reading Passage 3?

In boxes 36-40 on your answer sheet, write

YES if the statement is true

NO if the statement is false

NOT GIVEN if the information is not given in the passage

36 Two psychologists in Britain carried out an experiment which was open to question. ☐

37 Cooke used a technique of connection with a real object and a number in card. ☐

38 Simonides recalled each dead guest by each unique sound which belonged to every guest. ☐

39 Modern people's memory largely dropped because of relying too much on logic thinking. ☐

40 Ericsson gave his consent to the opinion of John Wilding and Eleanor Maguire who did earlier experiment. ☐

THE END